SFWRENG-3MX3 Assignment 5

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Contents

1) Impulse to Frequency Response

Use the Discrete Time Fourier Transform (DTFT) or the Continuous Time Fourier Series (CTFT) to find the frequency response that corresponds to the following impulse responses:

1.a) $h(t) = \delta(t-1) + 2\delta(t-2)$

Solution:

$$\begin{aligned} X(\omega) &= \int_{-\infty}^{\infty} (x(t)e^{-i\omega t})\, dt \\ &= \int_{-\infty}^{\infty} (\delta(t-1) + 2\delta(t-2)e^{-i\omega t})\, dt \\ &= \int_{-\infty}^{\infty} (\delta(t-1)e^{-i\omega t} + 2\delta(t-2)e^{-i\omega t})\, dt \\ &= e^{-i\omega} + 2e^{-2i\omega} \end{aligned}$$

1.b) $h(t) = \begin{cases} \frac{1}{3}, & \text{if } n = 0, 1, 2 \\ 0, & \text{otherwise} \end{cases}$

Solution:

2) The Step Response

In control theory, people use the step response in addition to the impulse response. The reason for this is that the step response gives information on how a controlled system reacts to a change of its set-point, which is a desired state for the system to be in. The step function is given by:

$$\text{step}(n) = \begin{cases} 0, & \text{if } n < 0 \\ 1, & \text{otherwise} \end{cases}$$

and the step response is the output of an LTI system in its zero state when $x(n) = \text{step}(n)$.

2.a) How might we compute the step response from the impulse response?

Solution:

2.b) Compute the step response of the system given by $y(n) = x(n) + \frac{1}{2}y(n-1)$ *directly.*

Solution:

2.c) Compute the step response of the system given by $y(n) = x(n) + \frac{1}{2}y(n-1)$ using the formula derived in **(a)**.

Solution:

3) Filter Design

Design a simple FIR filter that removes any frequency component at $\frac{\pi}{2}$ and passes any constant signal with a gain of 1. Give the difference equation and state space representations for such a filter.

Solution:

4) Filtering Signals

Given is a system that has the frequency response:

$$H(\omega) = \Big\{ 1, \text{ if } |w| \leq 2.5 \text{ radians/second}$$